Statistics Hybrid Practice Set – S12 (L8)

Topics Covered

- Population
- Sample
- Mean
- Median
- Mode
- Range
- Population Variance
- Population Standard Deviation
- Quartiles (Q1, Q2, Q3)
- IQR
- Outlier Detection
- Frequency Table
- Histogram
- **Density Plot (Introduction)** ⭐

Instructions

- Solve on **paper first**.
- Then solve the **same questions in Excel**.
- Explain your answers where required.

Q1. Identify Population and Sample

A hospital has **30,000 patients**. It randomly selects **1,000 patients** for a health survey.

Paper Task

- Population = ?
- Sample = ?

Excel Task

Total Patients	Surveyed Patients
30000	1000

Q2. Calculate the Mean

Value
16
20
24
28
32

Paper Task

- Calculate the Mean.

Excel Task

- Verify your answer.

Q3. Complete Statistical Analysis

Value
22
26
30
34
38

Paper Task

Calculate:

- Mean
- Range
- Population Variance
- Population Standard Deviation

Excel Task

Verify your answers.

Q4. Quartiles & IQR

Value
12
15
18
21
24
27
30
33
36

Paper Task

Find:

- Q1
- Q2
- Q3
- IQR

Excel Task

Verify your answers.

Q5. Outlier Detection

Value
40
42
44
46
48
50
52
54
110

Paper Task

Find:

- Q1
- Q3
- IQR
- Lower Fence
- Upper Fence

Is **110** an outlier?

Excel Task

Verify your answer.

Q6. Frequency Table

The following ages were recorded for **25 customers**.

Ages
18
21
23
24
26
28
31
33
35
36
38
41
43
45
47
49
52
54
56
58
61

Ages
63
66
69
72

Create a frequency table.

Age Group	Frequency
18–29	?
30–39	?
40–49	?
50–59	?
60–69	?
70–79	?

Excel Task

Create the table using COUNTIFS().

Q7. Histogram Interpretation

Using your frequency table from **Q6**:

Answer:

1. Which age group has the highest frequency?
2. Which has the lowest frequency?
3. Is the histogram approximately symmetric, left-skewed, or right-skewed?
4. Explain your answer in one sentence.

⭐ **Q8. Density Plot (Introduction)**

Two datasets contain **20 customer ages** each.

Dataset A

18, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 39

Dataset B

18, 18, 18, 19, 19, 20, 20, 21, 21, 22, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54

Answer:

1. Which dataset would have **one main peak**?
2. Which dataset would likely have **two peaks**?
3. Which dataset appears **more evenly distributed**?

Note: Do **not** create a density plot yet. Just interpret the data. We'll learn how to create one later.

Q9. Real Data Analyst Scenario

An HR manager wants to compare the **overall spread of employee salaries** between two departments.

Which statistical measure is the **best** choice?

A. Mode

B. Standard Deviation

C. Median

D. Frequency Table

Explain your answer in one sentence.

Q10. Mixed Business Case

A company recorded the following customer waiting times (minutes):

5, 6, 6, 7, 8, 9, 30

Answer:

1. Is there likely an outlier?
2. Which method would you use to confirm it?
3. Would you report the **Mean** or **Median** as the typical waiting time?
4. Explain your answer in one sentence.